

Probabilistic Models

Exam 2018WS

January 28, 2019

1. Bayesian Networks

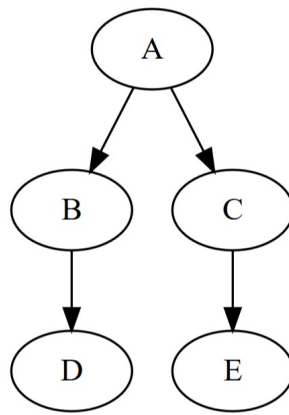


Figure 1: Bayesian Network

1. What is the Joint Probability Table of the given BN?
2. $B \perp D \mid A$?
3. How many *marginally* independent pairs $X \perp Y$?
4. Give examples for Reasoning Patterns:
 - Causal
 - Evidential
 - Intercausal

2. Sampling

1. What is the difference between Likelihood Weighting (LW) and Rejection Sampling (RS)?
2. What is the main problem with LW?
3. In what sense is LW an instance of Importance Sampling?
4. What is the main idea of Gibbs Sampling?

3. Model Fitting

	y_0	y_1
x_0	0.25	0.25
x_1	0.25	0.25

1. Give a BN representation where X and Y are independent.
2. Give a BN representation where $Y \rightarrow X$.
3. What does this tell us about the potential of model fitting?

4. State Observation Model

1. Why do we need the Markov Assumption (very restrictive)?
2. Why do we need the Stationary Assumption?

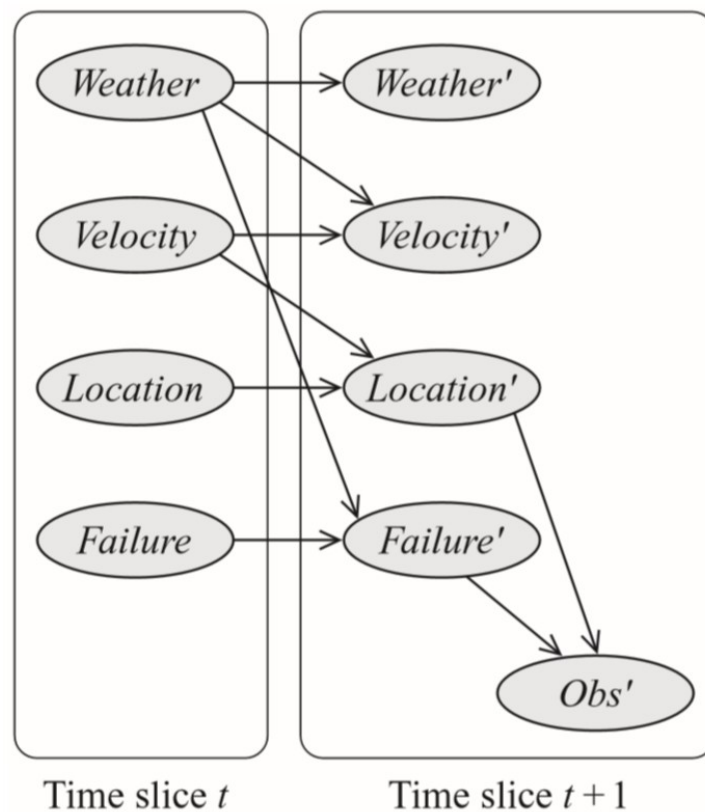


Figure 2: State Observation Diagram

3. Why is there an edge between $Weather$ and $Weather'$?
4. Is Obs' conditionally independent from $Weather$ ($P(Obs') = P(Obs' | Weather')$)?